

Requirements

AI-Based Fake Identity Document Screening System

System Prerequisites

Requirement	Minimum Version	Recommended	Purpose
Node.js	v16+	v18+ / v24 LTS	Backend proxy server
npm	v8+	v10+	Package management
Python	3.7+	3.10+	Frontend static file server
Web Browser	Chrome 80+	Latest Chrome/Edge	WebGL for TensorFlow.js

Backend Dependencies (Node.js)

Installed via "npm install" inside the server/ directory.

Package	Version	Purpose
express	^4.19.2	HTTP server and API routing
cors	^2.8.5	Cross-Origin Resource Sharing middleware
dotenv	^16.4.5	Load environment variables from .env file

Frontend Dependencies (CDN - No Install Required)

Loaded automatically via CDN <script> tags in index.html:

Library	Version	Purpose
TensorFlow.js	latest	ML inference engine in the browser
Teachable Machine (Image)	latest	Image classification model loader
Tesseract.js	v5	Client-side OCR text extraction
Font Awesome 6	6.4.0	UI icons
Inter Font	-	Typography

API Keys (Optional)

Key	Required?	Source	Purpose
GEMINI_API_KEY	Optional	Google AI Studio	AI chatbot for fake doc explanation

Note:

The core document screening (Aadhaar, PAN Card, Passport, Voter ID) works fully without a Gemini API key. The key is only needed for the AI Security Assistant chatbot.

Configuring the API Key

1. Navigate to server/.env
2. Replace the placeholder: GEMINI_API_KEY=your_actual_api_key_here
3. Restart the backend server

Pre-trained ML Models (Included)

These Teachable Machine models are bundled in the repository - no download needed:

Model Directory	Document Type	Files
model-aadhaar/	Aadhaar Card	model.json, weights.bin, metadata.json
model-pancard/	PAN Card	model.json, weights.bin, metadata.json
model-passport/	Passport	model.json, weights.bin, metadata.json
model-voterid/	Voter ID	model.json, weights.bin, metadata.json
model-doctype/	Document Type Classifier	model.json, weights.bin, metadata.json

Ports Used

Port	Service
5000	Frontend (Python HTTP server)
5001	Backend (Node.js Express API)

Ensure these ports are free before starting the application.

Quick Setup Commands

```
# 1. Install backend dependencies
cd server
npm install

# 2. Create .env file (copy from template)
cp .env.example .env
# Edit .env and add your GEMINI_API_KEY (optional)

# 3. Start backend server
npm start

# 4. In a new terminal - start frontend (from project root)
cd ..
python -m http.server 5000

# 5. Open in browser
# http://localhost:5000/login.html
```

Demo Credentials

Email: admin@example.com | Password: password123